



BFS in E^2

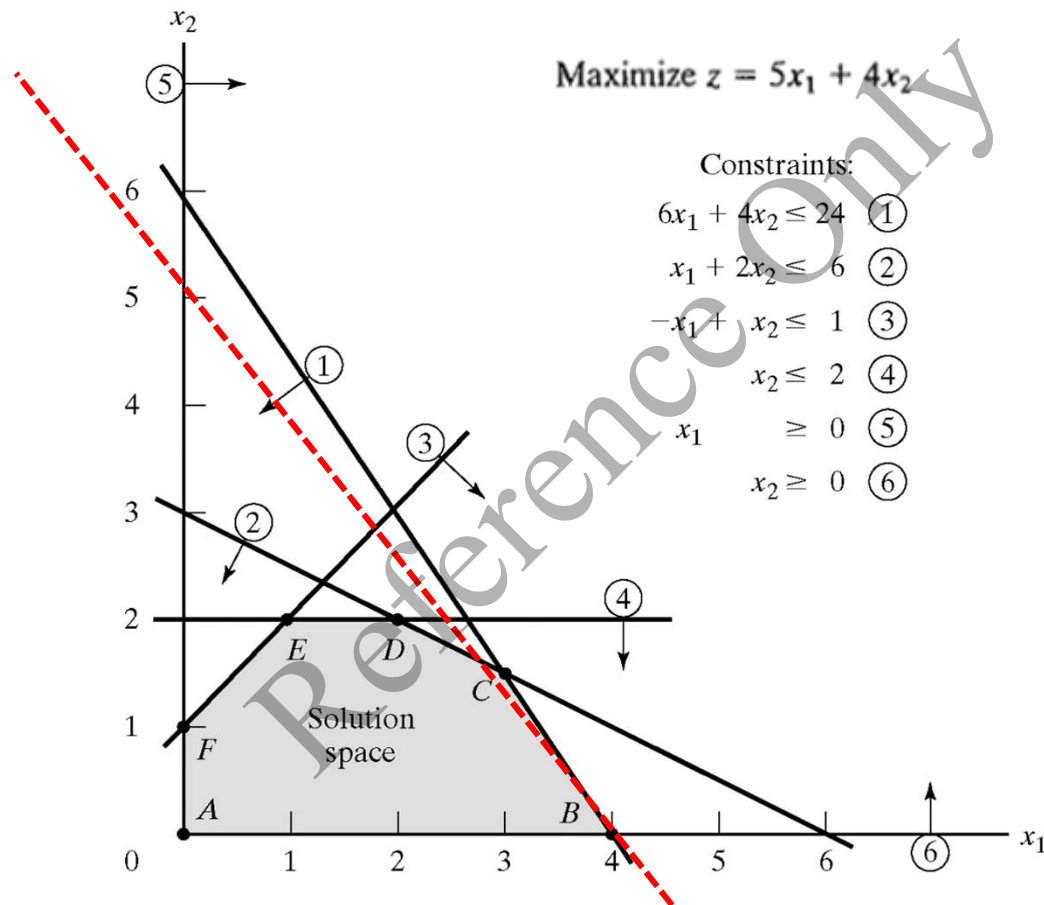


Figure 2.1

Feasible space of the Reddy Mikks model.



BFS in E^3

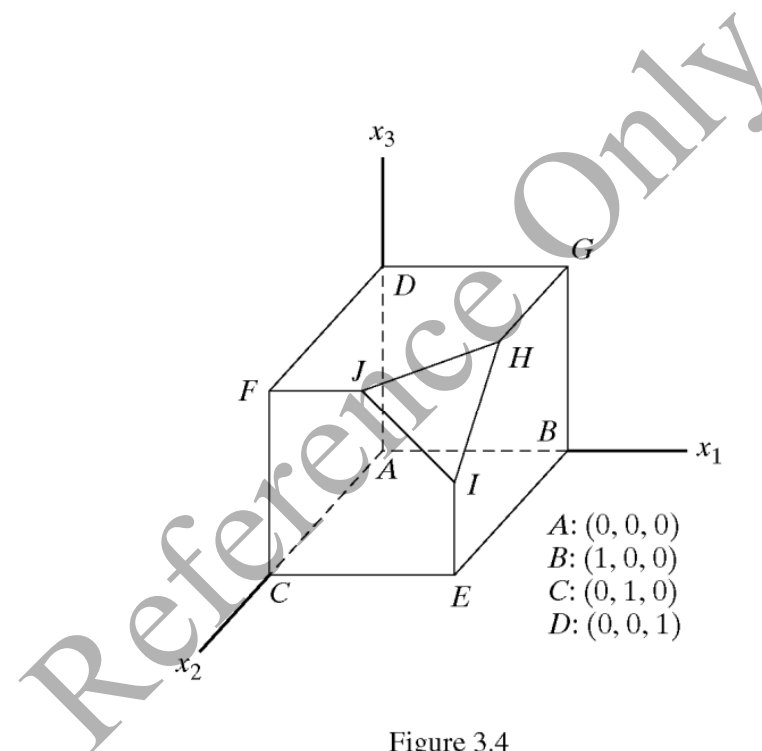
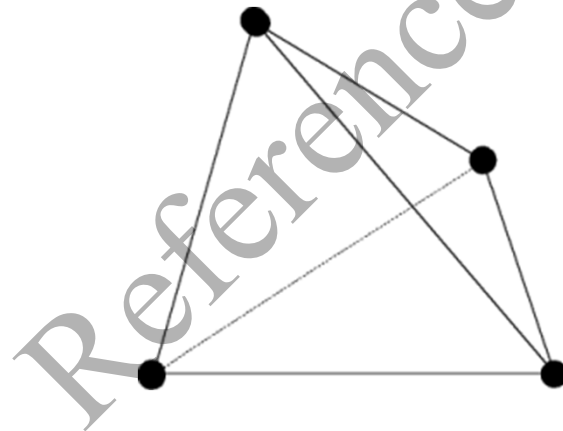


Figure 3.4
Solution space of Problem 3, Set 3.2b.



Simplex Defined

- A simplex is an n -dimensional **convex** polyhedron having exactly $(n+1)$ vertices.
- Simplices in various dimensions: $n=0$ is a point, $n=1$ is a line segment, $n=2$ is a triangle and $n=3$ is a tetrahedron.





The Simplex Algorithm

- Algorithm – is a step-by-step procedure for performing a computation.
- Simplex algorithm investigates only a “select few” of the BFSs.
- Simplex algorithm is an iterative procedure, i.e. a repetitive procedure.

Reference Only



The Algorithm in Simplices

- Examining the axis directions, the simplex algorithm is verifying whether the origin is optimal for this simplex or whether any of the other vertices of the simplex have a better objective value.
- The simplex method proceeds by examining one such simplex after another, each time inquiring whether the current vertex of the present simplex is optimal for this simplex or not.



Simplex Algorithm ver 1

1. Put the LP in standard form.
2. Get a starting BFS (the origin).
3. Test for Optimality
4. If optimal, STOP, else go to step 5. * (there are other branches for this step)
5. Determine the Entering Variable.
6. Determine the Leaving Variable.
7. Determine the new BFS.
8. Go to step 3.



Tableau Method

Maximize $z = 5x_1 + 4x_2$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



Maximize $z = 5x_1 + 4x_2$
subject to:

$$6x_1 + 4x_2 + s_1 = 24 \quad (1)$$

$$x_1 + 2x_2 + s_2 = 6 \quad (2)$$

$$-x_1 + x_2 + s_3 = 1 \quad (3)$$

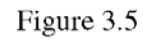
$$x_2 + s_4 = 2 \quad (4)$$

$$x_1, x_2 \geq 0$$

	x1	x2	s1	s2	s3	s4	Solution
z	-5	-4	0	0	0	0	0
s1	6	4	1	0	0	0	24
s2	1	2	0	1	0	0	6
s3	-1	1	0	0	1	0	1
s4	0	1	0	0	0	1	2



pivot row, pivot column, pivot element



L Lorenzo, UP IE/OR



Tableau Method

	x1	x2	s1	s2	s3	s4	Solution
z	0	-2/3	5/6	0	0	0	20
x1	1	2/3	1/6	0	0	0	4
s2	0	4/3	-1/6	1	0	0	2
s3	0	5/3	1/6	0	1	0	5
s4	0	1	0	0	0	1	2



Tableau Method

	x1	x2	s1	s2	s3	s4	Solution
z	0	0	3/4	1/2	0	0	21
x1	1	0	1/4	-1/2	0	0	3
x2	0	1	-1/8	3/4	0	0	3/2
s3	0	0	3/8	-5/4	1	0	5/2
s4	0	0	1/8	-3/4	0	1	1/2

Scarce and abundant resources

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$